

Last name	First name	Matriculation number



**University of
Zurich^{UZH}**

Computer Graphics **Sample Questions**

You have **90** Minutes time for solving this exam,

The following rules apply:

- The questions should be answered directly onto the exam sheets and their backside, clearly indicating questions and their answers. Additional sheets are distributed on request. Your name and student ID must be marked on each additional sheet used.
- Answers on additional paper will only be accepted if your **name** and **Matrikel-Nr.** are written on it, and the **question** you are answering is clearly indicated.
- Use a blue or black non-erasable pen to write down your answers. Do not use red; use only other colors where needed, e.g., in drawings and figures.
- The exam is “closed books”. No scripts or books are allowed. It is allowed to use an English dictionary. You will be allowed to take a one-sided A4 handwritten help sheet with you. You may write it by hand digitally and then print it, as long as it’s still handwritten (no handwriting-to-text conversion or so); see below for more details.
- It is allowed to use a simple non-programmable calculator without memory capacity.
- No other computing and electronic (communication) devices, such as computers, PDAs, smartphones, audio devices, etc., are allowed.
- You can use one single-sided sheet of A4 paper with your own handwritten notes. It has to be written personally by hand (no pencil or other erasable pens) and cannot be computer or typewriter-generated text or figures. Utilities to make the notes readable are prohibited (such as lenses for small-print fonts, color filters for multilayered notes, etc.). Notes that do not comply with these rules will be confiscated instantly, and the exam will have to be completed without any notes, with no exceptions.
- MC questions are scored on the Kprim principle: No points are given for 50% correct yes/no or true/false answers; from 50% to 100% correct answers, points are awarded proportionally.
- Please place your student ID card on the desk.

A. Vectors and Geometric Modeling (10 points)

1. (6 points) Considering a mesh that uses quadrilaterals (polygons with four corners) as primitives, name two possible pitfalls compared to a mesh with triangles as primitives. Explain how and where each problem would affect the standard real-time rendering pipeline.

Solution: Possible options (only two required):

- four vertices, unclear linear interpolation, affects interpolated vertex data (e.g. for shading and texturing) during rasterization and in the fragment shader
- non-convex, more difficult inside polygon test, affects rasterization
- non-planarity, no simple planar linear interpolation and thus no simple projection into image plane, affects rasterization, interpolation of vertex data, perspective projection of interior, etc. in rasterization etc.
- self-intersecting edges, unclear inside test and interpolation, affects rasterization etc.

2. (4 points) The plane π is defined by the point $Q(1, 1, 1)$ and the unit normal $\hat{\mathbf{n}} = \frac{1}{\sqrt{3}}(1, 1, 1)$.
- a. Write the plane in general form as a function of x, y and z .
 - b. Compute the signed distance from $R(2, 2, 0)$ to the plane.

Solution:

$$+2 \quad \hat{\mathbf{n}} \cdot (\mathbf{X} - \mathbf{Q}) = 0 \implies x + y + z = 3, x + y + z - 3 = 0, z = 3 - x - y \text{ or some other similar way of formulation}$$

$$+2 \quad d = \hat{\mathbf{n}} \cdot (\mathbf{R} - \mathbf{Q}) = \frac{1}{\sqrt{3}}(1 + 1 - 1) = \frac{1}{\sqrt{3}} > 0$$

B. Transformations and Compositions (30 points)

1. (5 points) When we transform an object like a cube, in certain situations, we want to preserve specific geometric features like conserving any parallel lines or sides. Show that an affine transformation $T(x) = Ax + b$ preserves parallel lines.

Solution: The same vector direction v defines parallel lines L_1 and L_2 :

$$L_1 : p_1 + t_1 v$$

$$L_2 : p_2 + t_2 v$$

Give the affine transformation $T(x) = Ax + b$ and applied to both lines we obtain:

$$L_3 : T(p_1 + t_1 v) = A(p_1 + t_1 v) + b = (Ap_1 + b) + t_1 Av$$

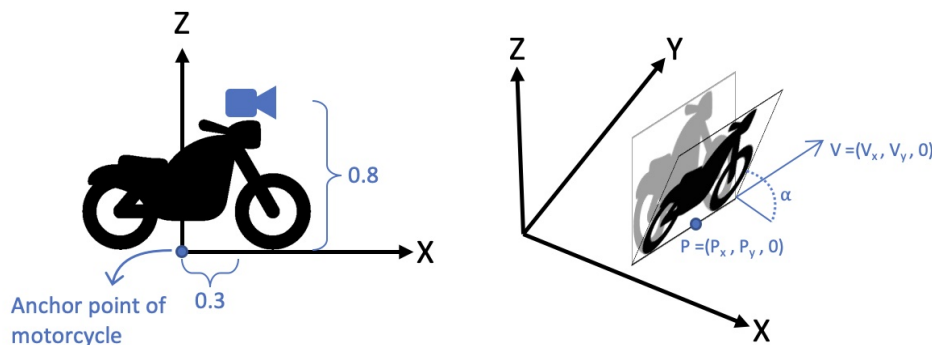
$$L_4 : T(p_2 + t_2 v) = A(p_2 + t_2 v) + b = (Ap_2 + b) + t_2 Av$$

Let $u = Av$; therefore, L_3 and L_4 are defined by the same direction vector u , so both transformed lines are parallel.

2. (25 points) Given an interactive motorcycle driving simulator, we are looking for the world to camera transformation matrix M . The desired camera view is attached to the middle of the handlebars of the motorcycle, and has a viewing direction along the motorcycle and parallel to the x, y ground plane at fixed height in z . In the motorcycle modeling coordinates, the camera's viewing axis is oriented with the motorcycle's x direction and has a vertical offset in z of 0.8 and a horizontal displacement of 0.3 in x with respect to the center of the motorcycle.

A driving simulator animates the motorcycle, and for each time t provides its current (center) position $P = (p_x, p_y, 0)$ and driving direction $V = (v_x, v_y, 0)$, as well as the motorcycle's lean-angle α in degrees. When the motorcycle is leaning completely to the right $\alpha = 0$, when standing upright $\alpha = 90$, and $\alpha = 180$ when leaning completely to the left.

Describe the construction and compute the world-to-camera transformation matrix M . Note that you can define what the principal viewing axis in the camera coordinate system should be, e.g. viewing direction x , left y and up z .



Solution: See explanations in the OLAT forum.

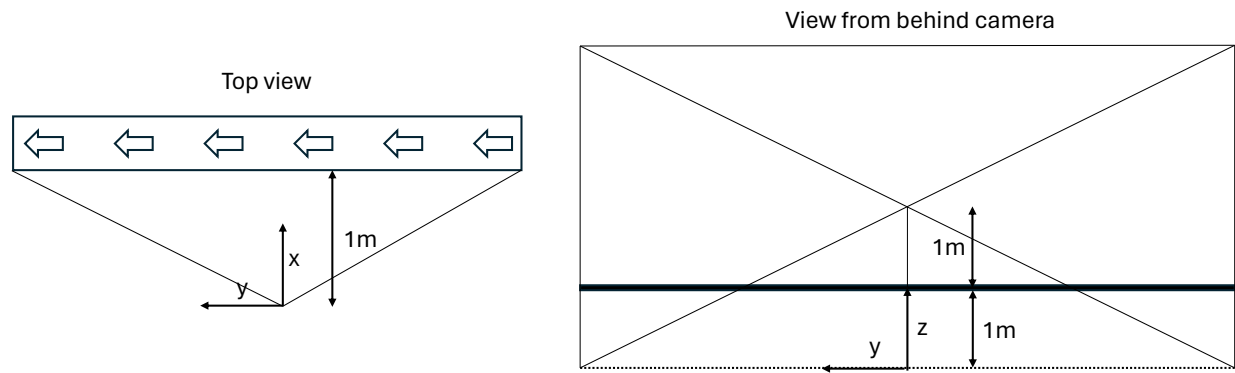
C. Viewings and Projections (15 points)

1. (15 points) A video camera is given to control products passing on a conveyor belt. The camera is mounted 1 Meter away from the conveyor belt and one meter above it. The conveyor belt is 1 meter above ground. The image plane is orthogonal to the ground. The camera records in 16 : 9 aspect ratio and is mounted so that it records horizontally.

Assume a local coordinate system with the origin below the camera with the x axis point towards the conveyor belt, the y axis pointing sideways and the z axis pointing up.

Describe and illustrate how to define a set of corresponding VRP, VPN, VUP, CW (clip window corners), and PRP parameters and compute their values. Indicate in which reference frame you have given the values of a particular parameter.

Sadly, after a while, the camera breaks and is replaced by a camera that records with a 4 : 3 aspect ratio (like old TVs). The camera mounting point cannot be moved but the new replacement camera should cover at least as much as the old one. How large must the vertical field of view of the new camera at least be? Explain which parameters additionally change and how they are changed.



Solution: Initial Setup:

$$VRP = (1, 0, 2)$$

$$VPN = (-1, 0, 0)$$

$$VUP = (0, 0, 1)$$

$$CW = [(1, -\frac{32}{9}, 0), (1, \frac{32}{9}, 0), (1, \frac{32}{9}, 4), (1, -\frac{32}{9}, 4)]$$

$$PRP = (0, 0, 2)$$

4:3:

$$CW = [(1, -\frac{32}{9}, -\frac{2}{3}), (1, \frac{32}{9}, -\frac{2}{3}), (1, \frac{32}{9}, \frac{14}{3}), (1, -\frac{32}{9}, \frac{14}{3})]$$

$$FOV = 2\arctan^{-1}(32/9) = 2.6(149^\circ)$$

D. Illumination, Shading, and Rasterization (2 points)

1. (2 points) A cube is illuminated by a single directional light coming from $\mathbf{L} = (1, 1, 1)$. Each cube face is drawn as an independent *quad* (by two triangles), and the normal of the front face is perfectly aligned with the light. Will one specific pixel on that face look *different* when rendered a) with *flat shading* and b) with *Phong shading*? Give a short justification.

Solution: No difference.

For this face all four per-vertex normals equal the face normal. Flat shading uses that face normal for every fragment;

Phong shading interpolates the *same* normal, so the lighting equation produces identical colour in both cases.

E. Texturing, Normal and Bump Mapping (4 points)

1. (4 points) Explain why area averaging is hardly used and present an alternative way to solve the aliasing problem.

Solution: Costly to find all covered texels, requiring many memory accesses

Instead one can use mip-mapping, hence create lower resolution textures by averaging, approximating the area averaging, usually a quarter of the resolution with each step

F. Visibility, Shadows, and Ambient Occlusion (2 points)

1. (2 points) To most directly reduce shadow *acne* when using basic depth-biased shadow maps, a common technique is to apply a _____ while rendering the shadow map. This adjusts the depth bias based on surface angle, helping to prevent self-shadowing artifacts.

Solution: slope-scaled depth bias

G. PBR and Ray Tracing (4 points)

1. (2 points) Which of the following is NOT a good argument for rasterization rather than ray tracing? (Single choice question)
 - ☐ A rasterizer does not have to store the entire scene in memory at once.
 - ☐ Modern graphics hardware support rasterization is supported.
 - ✓ **Rasterization provides order-independent transparency.**
 - ☐ In a ray tracer, traversal of spatial data structures can require highly incoherent memory accesses.
 - ☐ If the geometry changes, a ray tracer may need to rebuild a spatial data structure.
2. (2 points) Why do we use Monte Carlo methods to integrate the rendering equation rather than a more standard numerical quadrature like the trapezoid rule? (Single choice question)
 - ☐ To avoid aliasing.
 - ☐ Because standard quadrature is not uniform in area.
 - ✓ **Because of the curse of dimensionality.**
 - ☐ So that we can get global rather than just local illumination.
 - ☐ Because not all functions are piecewise linear.